

# NEET Important Chapter – Wave Optics

## Wave Optics an Important Concept for NEET

Search your topic here...

The chapter on Wave Optics starts with the basic revision of all the concepts we have studied in our lower grades to build a connection between the advanced parts of Wave Optics. In Wave Optics, we will discuss Huygens Principle, Reflection and Refraction of Plane Waves using Huygens Principle, Interference of Light Waves, Young's Double Slit Experiment, Coherent and Non-Coherent Sources, Diffraction, etc.

We will get introduced to a few most important concepts: Diffraction and Polarisation along with the polarisation by scattering and reflection.

In this chapter, students will also get to learn about the validity of Ray Optics and Resolving Power of Optical Instruments alongside the Doppler Effect.

Now, let us move on to the important concepts and formulae related to NEET and NEET exams along with a few solved examples.

## Physics Wave Optics Important Concepts for NEET

| Name of the Concept | Key Points of the Concept                                                                                                                                                                                                                                                                  |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Wavefront           | <ul style="list-style-type: none"> <li>• If all the particles vibrate in the same phase at a given instant then the locus of all those particles is called a wavefront.</li> <li>• This wavefront is divided into three categories: Spherical, Cylindrical and Plane wavefront.</li> </ul> |

|                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Huygen's Principle                | <ul style="list-style-type: none"> <li>• Huygen's Principle is mainly a geometrical construction based on the following points</li> <li>• Each point of the wavefront is a source of secondary disturbance which spreads in all directions.</li> <li>• Each point on primary wavelets produces secondary wavelets which travel in space with the speed of light.</li> <li>• The forward envelope of the secondary wavelets at any instant gives the new wavefront.</li> <li>• In a homogeneous medium, the wavefront is always perpendicular to the direction of wave propagation.</li> </ul> |
| Interference of Light             | <ul style="list-style-type: none"> <li>• The phenomenon of redistribution of light energy in a medium when the light waves from two coherent sources superimpose one another is called interference of light waves</li> <li>• Interference here is considered of two types:</li> <li>• Constructive and Destructive</li> </ul>                                                                                                                                                                                                                                                                |
| Coherent and Non-Coherent Sources | <ul style="list-style-type: none"> <li>• If two sources are having a constant phase difference between them, then those sources can be considered as coherent sources.</li> <li>• These two sources should give light of the same frequency (or wavelength).</li> </ul>                                                                                                                                                                                                                                                                                                                       |

|                            |                                                                                                                                                                                                                                                                                                                                                                    |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Principle of Superposition | <ul style="list-style-type: none"> <li>• This principle states that when two or more waves with the same frequency overlap, the resultant wave is the algebraic sum of individual waves.</li> </ul>                                                                                                                                                                |
| Diffraction of Light       | <ul style="list-style-type: none"> <li>• The phenomenon of bending of light around the corners when light waves are obstructed by an obstacle is called diffraction.</li> <li>• Diffraction is categorised into 2 types.</li> <li>• Fraunhofer</li> <li>• Fresnel</li> </ul>                                                                                       |
| Doppler's effect in Light  | <ul style="list-style-type: none"> <li>• The change in the frequency of source when either the observer moves towards the source or far from the source is called Doppler's Effect.</li> <li>• This is mainly of two types:</li> <li>• Red Shift (decreasing apparent frequency)</li> <li>• Blue Shift (increasing apparent frequency)</li> </ul>                  |
| Polarisation               | <ul style="list-style-type: none"> <li>• The phenomenon of confining the direction of wave vibration of an electric vector in one particular direction perpendicular to wave propagation's direction is called polarisation.</li> <li>• Polarisation is mainly of two types</li> <li>• Polarisation by Scattering</li> <li>• Polarisation by Reflection</li> </ul> |

| Sl. No | Name of the Concept                                | Formula |
|--------|----------------------------------------------------|---------|
| 1.     | Condition of maximum intensity in interference     | —       |
| 2.     | Condition of minimum intensity                     | —       |
| 3.     | Fringe Width in vacuum                             | —       |
| 4.     | Fringe width in any medium having refractive index | —       |
| 5.     | Resolving power of optical instruments             | —       |
| 6.     | Malus law                                          |         |
| 7.     | Brewster's law                                     |         |
| 8.     | Red Shift                                          | —       |
| 9.     | Blue Shift                                         | —       |

## Important Topics of NEET Wave Optics Chapter

### 1. Wavefront and Huygens Principle:

- Wave fronts are imaginary surfaces that connect all points of a wave that are in phase. They serve as a visualization tool to understand how waves propagate.
- Huygen's principle, on the other hand, postulates that each point on a wave front can be treated as a secondary source of spherical wavelets. These wavelets combine to create a new wave front.
- Together, wavefronts and Huygens principle are essential tools for comprehending the behavior of light and other wave phenomena.



## **2. Reflection and Refraction of a Plane Wave on a Plane Surface Using Wavefront:**

- When a wavefront encounters a reflecting or refracting surface, Huygen's principle comes into play. This principle states that each point on the wavefront acts as a secondary source of spherical wavelets.
- These wavelets combine to create a new wavefront after reflection or refraction. This process aids in predicting the behavior of light waves at interfaces, offering valuable insights into the phenomena of reflection and refraction.
- It's a foundational concept in wave optics, essential for understanding how light interacts with various surfaces.

## **3. Proof of Laws of Reflection and Refraction Using Huygens principle:**

- Huygens' principle is a powerful tool for proving the laws of reflection and refraction. For reflection, Huygens' principle explains that each point on an incident wavefront acts as a secondary source of spherical wavelets.
- These wavelets collectively construct the reflected wavefront. When applied to refraction, the principle clarifies the bending of wavefronts as they enter a different medium.
- The laws of reflection and refraction can be deduced by considering the specific geometry of these wavelets.
- This elegant approach not only mathematically confirms the laws but also provides a deeper insight into the behavior of waves at interfaces, making it an indispensable concept in the study of wave optics.

## **4. Interference:**

- Interference is a fascinating phenomenon that occurs when two or more waves overlap. It results from the superposition of these waves, leading to the formation of regions with constructive and destructive interference.
- In constructive interference, waves align in phase and amplify each other, creating areas of maximum intensity, known as bright fringes.
- In contrast, destructive interference occurs when waves are out of phase, causing them to cancel each other out in regions of minimum intensity, known as dark fringes.
- Interference plays a vital role in understanding the behavior of light, as it helps explain patterns observed in double-slit experiments and interference in thin films, providing valuable insights into the wave nature of light.

## 5. Young's Double Hole Experiment and Expression for Fringe Width:

Young's double-slit experiment is a fundamental demonstration of wave interference. It involves a screen with two closely spaced slits through which light passes. When the waves emerging from these slits overlap on a distant screen, they create an interference pattern. The expression for fringe width ( $W$ ) in Young's experiment is given by:

$$W = \frac{\lambda D}{d}$$

Where:

- $\lambda$  represents the wavelength of light used.
- $D$  is the distance between the double slits and the screen.
- $d$  is the separation between the two slits.

This equation reveals that fringe width is directly proportional to the wavelength and the distance between the screen and the slits while inversely proportional to the separation of the slits. Young's double-slit experiment is a crucial illustration of wave interference and provides insight into the behavior of light.

## 6. Coherent Sources and Sustained Interference of Light:

- Coherent sources are those that emit waves of the same frequency and with a constant phase difference. For sustained interference, it's crucial that the waves from these sources maintain their phase relationship.
- In practice, lasers and certain types of filtered light sources are often used to create coherent beams.
- When coherent waves overlap, they form a stable interference pattern, such as the well-known Young's double-slit experiment.
- Coherent sources enable us to observe and study interference phenomena with precision, providing valuable insights into the wave nature of light.

## 7. Diffraction Due to Single Slit:

- When a wave, such as light, passes through a narrow slit, it doesn't simply create a sharp, well-defined image on the other side. Instead, it spreads out and generates a pattern of alternating bright and dark regions.
- This pattern is characterized by a central maximum and multiple smaller side maxima and minima.
- The width of these maxima and minima depends on the width of the slit and the wavelength of the wave.
- Diffraction due to a single slit is a fundamental concept that highlights the wave nature of light and is crucial for understanding various optical phenomena.

## 8. Width of Central Maximum:

When a wave, such as light, passes through a narrow slit, it creates a central maximum at the center of the diffraction pattern. The width of this central maximum depends on two key factors: the wavelength of the wave ( $\lambda$ ) and the width of the slit ( $a$ ).

Mathematically, the width ( $w$ ) of the central maximum is given by:

—

This formula illustrates that the central maximum becomes narrower with a decrease in the wavelength of light and a broader slit width. Understanding the width of the central maximum is fundamental in analyzing and predicting the diffraction patterns produced by single slits and is a key concept in wave optics.

## 9. Resolving Power of Microscopes and Astronomical Telescopes:

Resolving power measures the ability of an optical instrument to distinguish between two closely spaced objects. For microscopes, it's essential in observing tiny structures, such as cells. In telescopes, it's crucial for distinguishing celestial objects.

The resolving power ( $R$ ) of an optical instrument is given by the Rayleigh criterion:

—

Here,  $\lambda$  is the wavelength of light, and  $D$  is the diameter of the objective lens or mirror. Smaller values of  $\lambda$  or larger values of  $D$  result in higher resolving power.



Astronomical telescopes use resolving power to distinguish distant stars and galaxies. Microscopes employ it to see fine details within small specimens, making it a fundamental concept in the field of wave optics.

## **10. Polarization:**

Polarized light waves vibrate in a specific direction, which can be vertical, horizontal, or any angle in between. This property is extensively used in various applications, including glare reduction in sunglasses, enhancing 3D movie experiences, and in scientific tools like polarimeters. Understanding polarization is crucial in wave optics as it allows for control and manipulation of light waves for diverse purposes.

## **11. Plane polarized light; Brewster's Law:**

- Plane-polarized light is a type of light in which the electric field vibrations occur in a single plane. This means that the light waves oscillate in one specific direction, typically horizontal or vertical, rather than in all directions as in unpolarized light.
- Plane-polarized light is crucial in many applications, from reducing glare on water surfaces to enhancing the quality of liquid crystal displays.
- Brewster's Law is an essential concept related to plane-polarized light. It states that when light is incident on a transparent medium at a specific angle known as the Brewster angle, the reflected light becomes completely polarized.
- Brewster's Law plays a significant role in optics, especially in the design and optimization of anti-glare coatings and polarizing filters for various optical instruments and devices.

Understanding these principles is fundamental in the context of wave optics in NEET Physics.

## **12. Uses of Plane-Polarized Light and Polaroids**

Plane-polarized light and polarizers find extensive use in a wide range of applications.

- **Glare Reduction:** Polarized sunglasses employ the properties of plane-polarized light to reduce glare from surfaces like water, roads, or car windows. This enhances visibility and safety, making them indispensable for outdoor activities.
- **3D Cinema and Television:** Plane-polarized light is crucial for creating the 3D effect in cinema and television. Special glasses with differently polarized lenses for each eye ensure that each eye receives a slightly different image, creating the illusion of depth and immersion.
- **Analytical Chemistry:** Polarimeters utilize plane-polarized light to determine the concentration and composition of optically active substances in solutions. This is essential in pharmaceutical and chemical industries.
- **Material Testing:** In materials science, polarized light is used to examine stress patterns and internal structures in transparent materials. This is valuable in quality control and research.
- **LCD Displays:** Liquid crystal displays (LCDs) employ polarizers to control the intensity and orientation of light, enabling the formation of images on screens. Polarized light is pivotal in modern displays, from smartphones to televisions.

## NEET Physics Wave Optics Chapter Solved Examples

**1. Laser light of wavelength 630 nm incident on a pair of slits produces an interference pattern in which bright fringes are separated by 8.1mm. A second light produces an interference pattern in which the fringes are separated by 7.2mm. Calculate the wavelength of the second light.**

**Sol:**

Given,  $\lambda_1 = 630\text{nm}$  and  $\Delta x_1 = 8.1\text{mm}$   
and  $\Delta x_2 = 7.2\text{mm}$

———— for same value of D and d

is directly proportional to

Hence

—

= —

So — = — and

**= 560nm.**

**Therefore, the wavelength of the second light is 560 nm.**

**Key point:** Here only the relationship between wavelength and fringe width should be known to students.

**2. Find the distance for which ray optics is a good approximation for an aperture of 4mm and wavelength 400 nm.**

**Sol:**

Given,  $a = 4\text{mm}$ ,  $\lambda = 400\text{nm}$

= —

This — is called Fresnel distance for which ray optics is a good approximation for aperture  $a$  and wavelength

= —

**= 40m**

**Key point:** Here only the relationship between aperture and wavelength should be known to students.

**Previous Year Questions from NEET Papers**

**1. A beam of light of wavelength 600 nm from a distant source falls on a single slit 1 mm wide and the resulting diffraction pattern is observed on a screen 2m away. The distance between the first and dark fringes on either side of the central bright fringe is**

**Sol:**

Given wavelength = 600 nm,  $d = 1 \text{ mm} = 10^{-3} \text{ m}$  and  $D = 2 \text{ m}$

Hence, the central fringe width = ———

= —————

**=2.4mm**

**The distance between the first and dark fringes on either side of the central bright fringe is 2.4mm**

**Trick:** Here the formula of diffraction in a single slit can be used directly.

**2. In a double slit experiment, when light of wavelength 400 nm was used, the angular width of the first minima formed on a screen placed 1m away was found to be . What will be the angular width of the first minima, if the entire experimental apparatus is immersed in water? (Refractive Index of water = 4/3)**

**Sol:**

Given wavelength = 400 nm, Refractive index of water = 4/3, angular width in air = .

Angular width in air = —

Angular width in water = ———

= —

= ———

=



**Therefore, the angular width of the first minima is .**

**Trick:** Here one can see the medium dependency on angular width and how the formula changes for that.

### Practice Questions

**1. Find the ratio of intensities at two points on a screen in Young's double slit experiment when waves from the two slits have path difference of (i) 0 and (ii) — (Ans: 0 and 2:1)**

**2. A parallel beam of light of wavelength 500 nm falls on a narrow slit and the resulting diffraction pattern is observed on a screen 1 metre away. It is observed that the first minimum is at a distance of 2.5mm from the centre of the screen, find the width of the slit. (Ans: 0.2mm)**

### Conclusion

The study of Wave Optics is pivotal for NEET preparation. This chapter delves into the intricate behavior of light waves, explaining how they interact with various materials and influence phenomena like diffraction, interference, and polarization. A strong grasp of these concepts is essential for a comprehensive understanding of optics. Wave Optics finds practical applications in fields such as medical imaging and microscopy. Proficiency in this chapter not only enhances your performance in NEET but also provides valuable insights into the fascinating realm of light and waves, enriching your scientific knowledge. Learn with Vedantu's NEET Important Chapter Wave Optics and score well in the [NEET 2024](#) Examination.

Hi Aditya,

## Best courses for you

**Full syllabus LIVE  
courses**

Starting from ₹32,400

**One-to-one LIVE  
classes**

Starting from ₹0/hr

**Watch videos on**  
NEET Important Chapter – Wave Optics

Wave Optics Class 12 | One Shot | Marathon | JEE Main | JEE Advanced |Suri Sir| VJEE

**Subscribe**

Share

1.8K likes 72.1K Views 3 years ago

## Book your **Free Demo** session

Get a flavour of LIVE classes here at Vedantu

[Book a free demo](#)

## Vedantu Improvement Promise

We promise improvement in marks or get your fees back. T&C Apply\*